

Touch Through Sound: Ring-Mounted Auditory Pseudo-Haptics for Material Perception

Zhengte Cai
School of Future Technology
South China University of Technology
Guangzhou, Guangdong, China
ftztcai@mail.scut.edu.cn

Chiyue Wang
School of Future Technology
South China University of Technology
Guangzhou, Guangdong, China
202420163875@mail.scut.edu.cn

Yumiao Wang
Computer Science and Engineering
Tianjin University of Technology
Tianjin, China
dear3water@gmail.com

Jinghe Yu
School of Future Technology
South China University of Technology
Guangzhou, Guangdong, China
202364800522@mail.scut.edu.cn

Chen Li
Computer Science and Engineering
Tianjin University of Technology
Tianjin, China
cli@email.tjut.edu.cn

Zhanpeng Jin
School of Future Technology
South China University of Technology
Guangzhou, Guangdong, China
zjin@scut.edu.cn

Yang Gao*
School of Future Technology
South China University of Technology
Guangzhou, China
gaoyang@cs.ecnu.edu.cn

Abstract

We present Touch Through Sound, a ring-mounted auditory pseudo-haptics technique for material perception during real touch. Our method parameterizes texture cues (e.g., roughness and granularity) and continuously modulates contact sound from interaction dynamics (speed and a pressure proxy), while a ring-mounted speaker keeps audio co-located with the fingertip. In a within-subject study (N=6), our action-contingent synthesis outperformed a static playback baseline in perceived action-sound congruency, and ring-mounted output improved spatial congruency over headphones. Overall, combining action contingency and near-finger audio yielded the most realistic material impressions.

CCS Concepts

• Human-centered computing → Interaction techniques.

Keywords

Sound Feedback, Ring, Pseudo-Haptics, VR

ACM Reference Format:

Zhengte Cai, Chiyue Wang, Yumiao Wang, Jinghe Yu, Chen Li, Zhanpeng Jin, and Yang Gao. 2026. Touch Through Sound: Ring-Mounted Auditory Pseudo-Haptics for Material Perception. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3772363.3798554>

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798554>

1 Introduction

People judge surface materials through touch, yet high-fidelity haptic rendering typically requires specialized hardware that is difficult to scale beyond lab settings [1]. Pseudo-haptics offers an alternative by leveraging perceptual illusions to bias tactile experience without changing the physical surface [11, 16]. While prior work has largely emphasized vision, sound is tightly coupled with physical contact in everyday interactions and can provide a lightweight channel to augment touch [4, 8, 10, 13].

However, interactive audio is often layered on as discrete, event-triggered effects or is only weakly tied to users' continuous actions, limiting its effectiveness as pseudo-haptics [5, 18]. We highlight two gaps. First, sound is rarely treated as a controllable representation aligned with tactile-relevant perceptual dimensions. Second, many systems fail to preserve action and spatial contingency: in natural touch, contact sound varies continuously and with low latency as actions unfold, and its apparent origin is expected to match the point of contact [3, 6]. When audio violates these sensorimotor expectations in timing or spatial origin, interactions can feel detached and less plausible [2, 3].

Therefore, we design a ring-output auditory pseudo-haptics approach that (i) parameterizes tactile-relevant contact-sound cues and (ii) modulates them in real time with an action- and spatial-contingent mapping from touch dynamics (e.g., motion and pressure proxies) and contact location [17]. Rather than replaying material-specific recordings, we treat contact sound as a set of perceptually salient cues that can be tuned and combined, making audio a controllable design space for shaping touch [7, 8, 13, 14]. To support spatial congruency, we deliver audio near the finger via a ring-mounted output device. When visual context is used (e.g., in VR), it serves as an anchor and experimental control rather than the primary manipulation.

We implement an end-to-end prototype that continuously updates contact-sound cues during real touch and aligns the apparent source with the point of contact via near-finger output. We evaluate the approach with a study design that tests whether (1) tuning the parameterized contact-sound cues can systematically bias material-related judgments on the same physical surface, (2) action-contingent mappings improve perceived action–feedback coupling compared to event-triggered or single-signal baselines, and (3) near-finger output improves perceived spatial congruency compared to external audio output.

This work offers two contributions:

- (1) A sound-first auditory pseudo-haptics approach that treats contact-sound cues as controllable parameters and binds them to real-time, action- and spatial-contingent mappings during real touch.
- (2) A working ring-output prototype and study design (with baselines and measures) for evaluating how parameterized cues, action contingency, and spatial congruency shape perceived touch and material-related judgments.

2 Related Work

Pseudo-haptics biases touch perception through crossmodal conflicts without physically changing a surface [11, 16]. While many techniques emphasize visual manipulation, classic audiotactile effects (e.g., the parchment-skin illusion) demonstrate that contact sound alone can systematically shift tactile sensation [4, 8]. Building on this foundation, recent HCI systems have explored audio-tactile material and texture experiences via sonification or combined haptic–audio feedback [14, 18]. Perceptual studies further show that touch-produced sound contributes to roughness judgments and can be tuned to modulate perceived roughness [9, 13]. However, many implementations rely on material-specific playback, which limits systematic control over texture-relevant auditory cues. Moreover, for audio to function as a convincing pseudo-haptic channel, feedback must remain action-contingent and plausible in timing and spatial origin [2, 3]. In this work, we treat sound as a primary control channel for material perception during real touch by parameterizing auditory cues to enable controlled modulation of perceived roughness and granularity, using a continuously updating mapping tied to interaction dynamics, and delivering sound near the contact point via ring-mounted audio.

3 Method

We translate our motivation into three design goals (DG1–DG3) and implement them via parameterized texture cues, continuous action-contingent modulation, and ring-mounted audio for spatial co-location.

3.1 Design Goals

Our goal is to use sound as a controllable pseudo-haptic channel for material perception during real touch. To be convincing, the auditory feedback must not only encode texture-related cues, but also preserve the sensorimotor expectations of touch: it should be continuously driven by the user’s ongoing interaction and appear to originate from the contact point. [2, 3]

- (1) **DG1: Texture cue controllability.** A small set of parameters should produce distinguishable and tunable material impressions along tactile-relevant dimensions (e.g., roughness and granularity) without requiring per-material recordings.
- (2) **DG2: Action contingency.** Audio should respond continuously and with low latency to interaction dynamics (e.g., speed, pressure, press rate), rather than only triggering discrete sound events.
- (3) **DG3: Spatial congruency.** Audio should be perceived as originating near the contact point and remain consistent as the finger moves, matching the “touch here, sound here” prior.

3.2 Texture Parameterization and Action-Contingent Modulation (DG1, DG2)

3.2.1 Texture Cues as Structural Invariants. We do not aim to replicate a material’s “true” sound. Instead, we represent materials through a low-dimensional set of auditory cues that are relevant to tactile judgments and can be tuned and combined. [6, 15] In this work we focus on two example texture dimensions: *roughness* and *granularity*. Each action type k uses a compact parameter vector $m_k \in [0, 1]^d$ that controls a family of contact-sound timbres for that action. Treating m_k as structural invariants clarifies the independent variables in our evaluation: cue settings can be changed while holding the physical surface and action constant. This realizes DG1 by making texture impressions a low-dimensional, tunable cue space independent of the physical surface and action.

3.2.2 Action-Contingent Modulation as Transformational Invariants. While texture cues define “what it feels like,” sensorimotor plausibility depends on “how it reacts” to interaction. Event-based playback is insufficient because the same surface contact can vary continuously with speed, force, and action transitions. [12] Similarly, a single action variable (e.g., speed-only) is often under-specified for explaining tactile impressions across rubbing, pressing, and tapping.

We therefore extract action features $a_k(t)$ from touch and use them to modulate the audio continuously. The mapping is action-specific but follows a shared structure: a texture-dependent base plus action-driven excitation/modulation. This formulation supports replication of the idea without requiring our exact DSP implementation: another system can instantiate $\text{Material}_k(\cdot)$ and $\text{Action}_k(\cdot)$ differently while preserving the same control structure and measured variables. This mapping targets DG2 by ensuring continuous, low-latency action–sound coupling during touch.

3.2.3 Action-Specific Instantiations. For **rub** action, we use tangential speed $v(t)$ and a contact-strength proxy $p(t)$. Speed modulates perceived “scratchiness” and micro-event density (e.g., denser/brighter cues at higher v), while contact strength primarily modulates loudness/energy. Texture parameters in m_{rub} control roughness- and granularity-related timbral cues that remain stable across changes in $v(t)$ and $p(t)$.

For **press** action, we use a contact-strength proxy $p(t)$ and press rate dp/dt . Press rate modulates the amount of transient micro-activity during loading (more “crunch” when pressing faster), while maintaining low activity during sustained holding to match the

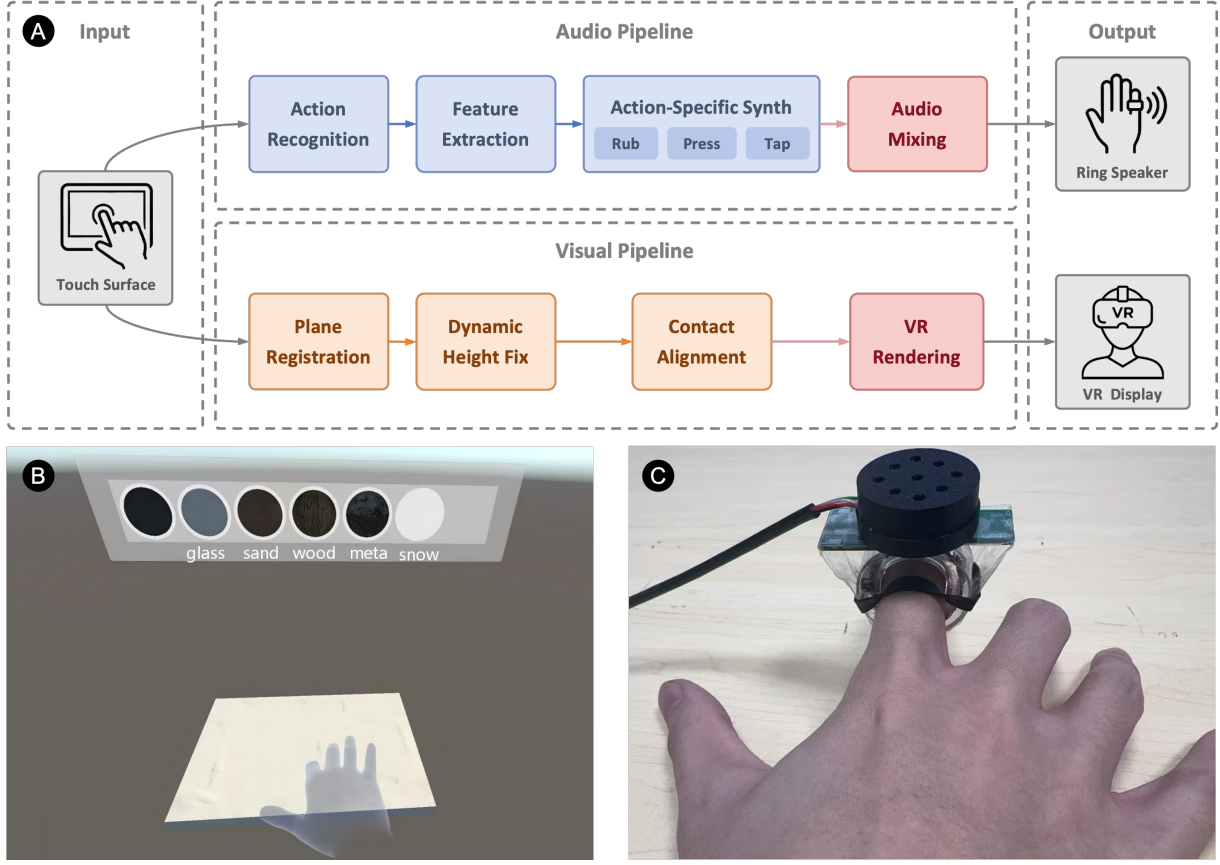


Figure 1: Overview of the prototype system implementation. (A) The system architecture uses a shared touch event stream to drive two parallel pipelines: the *audio pipeline* performs real-time action recognition and synthesis for rub, press, and tap interactions; the *visual pipeline* ensures spatiotemporal congruency through plane registration and dynamic height correction. **(B)** The VR environment provides visual context, allowing users to interact with different materials (e.g., glass, sand, wood, metal, snow) on a virtual tabletop aligned with the physical surface. **(C)** A custom ring-mounted speaker prototype co-locates the sound source with the fingertip to maintain spatial congruency during exploration.

quasi-static nature of pressing. Texture parameters in m_{press} control the character of these micro-activities (e.g., fine vs. coarse granularity) independently of how quickly the user presses.

For **tap** action, we treat tapping as a discrete impact characterized by peak contact strength p_{peak} , a sharpness proxy (peak dp/dt), and contact duration contact_s . These features modulate impact intensity, crispness/brightness, and effective damping, supporting controllable variations along hardness- and damping-related impressions via m_{tap} .

3.3 Spatial Congruency via Ring Output (DG3)

This section implements DG3 by making source co-location a controllable factor. Spatial congruency is a key condition for credible pseudo-haptics because users expect contact sound to originate near the point of touch. We therefore make source location an explicit, controllable factor in the method. In our prototype we achieve this by presenting audio from a ring-mounted device near the finger, but the method generalizes to any technique that can align perceived source location with contact location (e.g., near-finger speakers or other localized output).

4 System Implementation

Our system is anchored on a shared stream of touch events from an instrumented touch surface (touchpad), providing timestamped contact position and a pressure proxy. This stream drives two parallel pipelines: an *audio pipeline* for action recognition, feature extraction, and action-specific synthesis of pseudo-haptic contact sounds, and a *visual pipeline* that renders a VR tabletop aligned to the physical surface to maintain spatiotemporal congruency between what users see, feel, and hear.

4.1 Audio Pipeline

The audio pipeline implements the method described in Section 3, without introducing concepts beyond action features and action-specific synthesizers. Incoming touch events are treated as a continuous control stream; accordingly, action features are computed online to update synthesis parameters dynamically as the user’s action unfolds.

Action recognition and gating. We distinguish rubbing, pressing, and tapping on the same input stream using instantaneous per-event gating rules. Rubbing is activated when tangential motion exceeds a minimum velocity threshold (≥ 0.3 cm/s). Pressing is activated during quasi-static contact in which pressure changes but motion remains small (≤ 1.0 cm distance). Tapping is detected as a discrete impact when both pressure and pressure rise rate (dp/dt) exceed their thresholds within a brief time window (≤ 300 ms); if not, a fallback check fires on the lift event using peak pressure and total contact duration (≤ 220 ms). Tap detection is evaluated first on each incoming event to avoid double-triggering with presses.

Feature extraction and update rates. For each active touch, we compute tangential speed $v(t)$ from successive positions, while the press rate dp/dt is derived from successive pressure samples after light smoothing (30 ms low-pass time constant) to mitigate sensor noise. We update these features event-by-event, synchronous with the input sampling rate (100–150 Hz). The synthesizers read the latest feature values when producing audio blocks. Our audio engine runs at 44.1 kHz with 256-sample buffers (≈ 5.8 ms per buffer). This corresponds to an audio callback rate of about 172 Hz. Thus, feature changes affect output on the next audio callback. We smooth parameter changes over a short time window to reduce zipper noise while maintaining responsiveness. Overall, the end-to-end latency from physical action to auditory output depends primarily on the input sampling interval and audio buffer (≤ 16 ms for rubbing and early-trigger taps), while the dp/dt smoothing filter adds ≈ 30 ms of group delay for pressing.

Audio mixing. All three action synthesizers (rub/press/tap) render mono audio that is mixed and peak-limited for safety. We play back the output in real time to the ring-mounted speaker. This keeps the source near the finger and supports spatial congruency without explicit DSP.

4.2 Ring Prototype

The purpose of the ring prototype is to keep the apparent sound source near the point of contact. Because the speaker is worn on the hand, the source moves with the user's finger during exploration. It does not remain fixed in the environment. This physical co-location reduces ambiguity about whether the sound is caused by the user's current touch. It also provides a consistent spatial reference across changes in hand position. In our studies, ring output implements spatial congruency as a controllable factor. External audio output serves as a baseline that preserves the same synthesis but breaks co-location.

4.3 VR Alignment

Visual context can strengthen or undermine pseudo-haptics depending on whether it remains consistent with the touch-audio loop; thus, we use VR primarily for context and alignment rather than as an additional manipulation. We co-register a virtual tabletop with the user's real tabletop in a mixed-reality setup to establish a shared reference plane, then apply a brief manual translation so the virtual tabletop coincides with the physical touchpad region. To reduce residual height error from hand tracking and scene reconstruction, we perform dynamic height correction: upon each detected physical

contact event, we adjust the virtual tabletop height to the current virtual fingertip height so the virtual hand visually "collides" with the surface at the same moment the user touches it.

5 User Study

5.1 Participants and Procedure

We conducted a within-subject user study with 6 participants (mean age = 22.7 years), all of whom reported normal hearing and prior experience with virtual reality systems. Participants wore a Pico 4 Ultra headset and interacted with virtual materials using natural touch gestures (e.g., sliding or tapping). Auditory feedback was delivered either through headphones or a ring-mounted speaker positioned near the finger. Each trial corresponded to one material; we tested five materials: snow, sand, wood, metal, and glass. During the interaction, the material's visual appearance was initially occluded, encouraging participants to rely on touch and sound. Participants then experienced two audio alternatives (A/B) in randomized order while performing the same predefined action. In the proposed condition (A), sound was continuously modulated by interaction dynamics (e.g., sliding speed and a pressure proxy), whereas the baseline (B) played a fixed, material-matched sound that did not vary with movement speed or pressure (i.e., non-action-contingent).

5.2 Measures

Participants rated both audio alternatives (A/B) on three perceptual dimensions using 7-point Likert scales. First, participants rated how well each sound matched their intuition of how sound should change with their actions (e.g., faster vs. slower movement, lighter vs. heavier contact), reflecting *action-sound congruency*. Second, they rated the extent to which each sound felt as if it originated from the point of contact between the finger and the surface, reflecting *spatial congruency*. Finally, after the material was revealed, participants rated the overall realism of each sound, reflecting how well the sound matched the material when considering sound, action, and surface together.

5.3 Results

Figure 2 summarizes ratings under a 2×2 within-subject design: *Audio* (action-contingent synthesis vs. static playback baseline) $\times$ *Output* (ring-mounted near-finger speaker vs. headphones). We report three dimensions aligned with our design goals: action-sound congruency (DG2), spatial congruency (DG3), and overall realism (DG2+DG3).

Action-sound congruency (DG2). Across all five materials, participants rated our action-contingent synthesis higher than the static baseline, indicating that speed/pressure-driven modulation better matched their interaction dynamics. The effect was consistent across materials, with slightly lower congruency for hard-smooth textures (e.g., glass).

Spatial congruency (DG3). Ring-mounted output increased perceived co-location of sound with the touch point compared to headphones for every material, supporting the importance of near-finger audio for spatial plausibility.

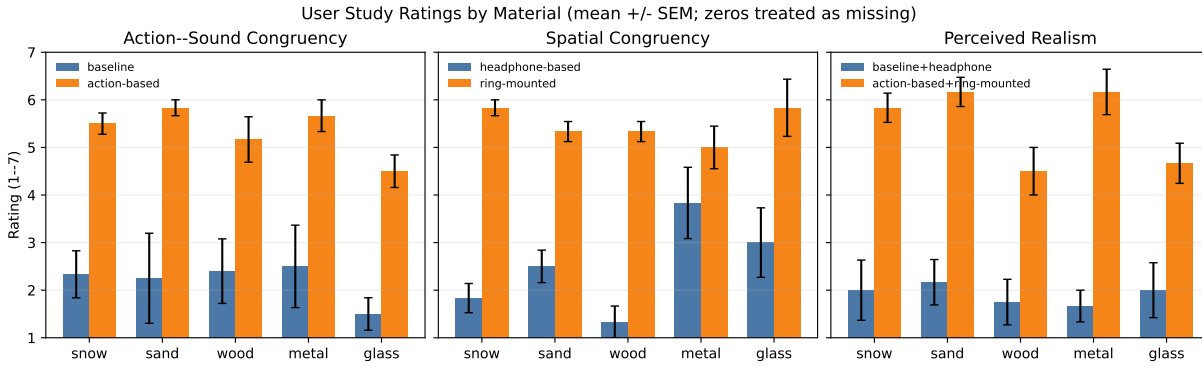


Figure 2: User study results (N=6) across five materials under a 2 $\times$ 2 design (Audio: action-contingent vs. static; Output: ring vs. headphones). Bars show mean $\pm$ SEM ratings for action-sound congruency (DG2), spatial congruency (DG3), and overall realism (DG2+DG3).

Material realism (DG1+DG2+DG3). Material realism was highest when combining action-contingent synthesis with ring-mounted output and lowest for the static baseline with headphones. This suggests complementary benefits of action contingency (DG2) and fingertip-level co-location (DG3) for a coherent touch-sound experience; moreover, the gain in material realism indicates that our parameterized cue space can produce plausible, material-specific impressions rather than generic contact sounds (DG1).

6 Conclusion

We introduced *Touch Through Sound*, a ring-mounted auditory pseudo-haptics approach for material perception during real touch. By parameterizing texture-relevant audio cues and continuously modulating contact sound from interaction dynamics, our method strengthens the perceived causal link between action and sound, while near-finger ring output improves the perceived co-location of sound with the touch point. A within-subject study across five materials suggests these factors provide complementary benefits, yielding the most realistic impressions when combined. Looking ahead, inspired by [17], we aim to move beyond touchpad-based sensing by leveraging wrist EMG together with ring IMU to predict contact force/pressure during everyday hand-object interactions, enabling fully on-device auditory pseudo-haptics without instrumented surfaces.

References

- [1] Cagatay Basdogan, Frederic Giraud, Vincent Levesque, and Seungmoon Choi. 2020. A Review of Surface Haptics: Enabling Tactile Effects on Touch Surfaces. *IEEE Transactions on Haptics* 13, 3 (July 2020), 450–470. doi:10.1109/TOH.2020.2990712
- [2] Detjon Brahimaj, Mondher Ouari, Anis Kaci, Frédéric Giraud, Christophe Giraud-Audine, and Betty Semail. 2023. Temporal Detection Threshold of Audio-Tactile Delays With Virtual Button. *IEEE Transactions on Haptics* 16, 4 (Oct. 2023), 491–496. doi:10.1109/TOH.2023.3268842
- [3] Elisa Canzoneri, Elisa Magosso, and Andrea Serino. 2012. Dynamic Sounds Capture the Boundaries of Peripersonal Space Representation in Humans. *PLoS ONE* 7, 9 (Sept. 2012), e44306. doi:10.1371/journal.pone.0044306
- [4] François Champoux, Olivier Collignon, Benoit A. Bacon, Franco Lepore, Robert J. Zatorre, and Hugo Théoret. 2010. Early- and Late-Onset Blindness Both Curb Audiotactile Integration on the Parchment-Skin Illusion. *Psychological Science* 22, 1 (Dec. 2010), 19–25. doi:10.1177/0956797610391099
- [5] Sonny Chan, Chase Tymms, and Nicholas Colonnese. 2021. Hasti: Haptic and Audio Synthesis for Texture Interactions. In *2021 IEEE World Haptics Conference (WHC)*. IEEE, Piscataway, NJ, USA, 733–738. doi:10.1109/WHC49131.2021.9517177
- [6] Alexis W. M. Devillard, Aruna Ramasamy, Xiaoxiao Cheng, Damien Faux, and Etienne Burdet. 2025. Tactile, Audio, and Visual Dataset During Bare Finger Interaction with Textured Surfaces. *Scientific Data* 12, 1 (March 2025), 46. doi:10.1038/s41597-025-04670-0
- [7] Nicola Di Stefano and Charles Spence. 2022. Roughness perception: A multisensory/crossmodal perspective. *Attention, Perception, & Psychophysics* 84, 7 (Aug. 2022), 2087–2114. doi:10.3758/s13414-022-02550-y
- [8] Veikko Jousmäki and Riitta Hari. 1998. Parchment-skin illusion: sound-biased touch. *Current Biology* 8, 6 (March 1998), R190–R191. doi:10.1016/S0960-9822(98)70120-4
- [9] Seung-Chan Kim, Ki-Uk Kyung, and Dong-Soo Kwon. 2007. The Effect of Sound on Haptic Perception. In *Second Joint EuroHaptics Conference and Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems (WHC'07)*. IEEE, Piscataway, NJ, USA, 354–360. doi:10.1109/WHC.2007.110
- [10] Norimichi Kitagawa and Charles Spence. 2006. Audiotactile multisensory interactions in human information processing. *Japanese Psychological Research* 48, 3 (Sept. 2006), 158–173. doi:10.1111/j.1468-5884.2006.00317.x
- [11] Marco Kurzweg, Yannick Weiss, Marc O. Ernst, Albrecht Schmidt, and Katrin Wolf. 2024. Survey on Haptic Feedback through Sensory Illusions in Interactive Systems. *Comput. Surveys* 56, 8 (April 2024), 1–39. doi:10.1145/3648353
- [12] C. Landelle, J. Danna, B. Nazarian, M. Amberg, F. Giraud, L. Pruvost, R. Kronland-Martinot, S. Ystad, M. Aramaki, and Anne Kavounoudias. 2021. The impact of movement sonification on haptic perception changes with aging. *Scientific Reports* 11 (2021), 5124. doi:10.1038/s41598-021-84581-3
- [13] Susan J. Lederman, Roberta L. Klatzky, Tamara L. Morgan, and Cheryl Hamilton. 2002. Integrating multimodal information about surface texture via a probe: relative contributions of haptic and touch-produced sound sources. In *Proceedings 10th Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems. HAPTICS 2002*. IEEE Computer Society, Washington, DC, USA, 97–104. doi:10.1109/HAPTIC.2002.998946
- [14] Roberto Montano-Murillo, Dario Pittera, William Frier, Orestis Georgiou, Marianna Obrist, and Patricia Cornelio. 2024. It Sounds Cool: Exploring Sonification of Mid-Air Haptic Textures Exploration on Texture Judgments, Body Perception, and Motor Behaviour. *IEEE Transactions on Haptics* 17, 2 (April 2024), 237–248. doi:10.1109/TOH.2023.3320492
- [15] Zhimin Ren, Hengchun Yeh, and Ming C. Lin. 2010. Synthesizing Contact Sounds Between Textured Models. In *2010 IEEE Virtual Reality Conference (VR)*. IEEE, Piscataway, NJ, USA, 139–146. doi:10.1109/VR.2010.5444799
- [16] Yusuke Ujitoko and Yuki Ban. 2021. Survey of Pseudo-Haptics: Haptic Feedback Design and Application Proposals. *IEEE Transactions on Haptics* 14, 4 (Oct. 2021), 699–711. doi:10.1109/TOH.2021.3077619
- [17] Yingjing Xiao, Zhichao Huang, Junbin Ren, Yuting Bai, Haichuan Song, Zhanpeng Jin, and Yang Gao. 2025. Wrist2Finger: Sensing Fingertip Force for Force-Aware Hand Interaction with a Ring-Watch Wearable. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*. Association for Computing Machinery, New York, NY, USA, 24:1–24:16. doi:10.1145/3746059.3747767
- [18] Jing Xue, Roberto Montano Murillo, Christopher Dawes, William Frier, Patricia Cornelio, and Marianna Obrist. 2024. FabSound: Audio-Tactile and Affective Fabric Experiences Through Mid-air Haptics. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 1–17. doi:10.1145/3613904.3642533

A Action-Specific Sound Models

This appendix details synthesis models for rubbing, pressing, and tapping, optimized for real-time control via material parameters and interaction dynamics.

A.1 Notation and Normalization

Inputs include contact position (x, y) and pressure $p(t)$. Speed $v(t)$ and press rate $\dot{p}$ are estimated from successive positions and smoothed pressure. Inputs are normalized to $[0, 1]$:

$$\hat{v} = \text{clip}\left(\frac{v}{v_{\max}}, 0, 1\right), \quad \hat{p} = \text{clip}\left(\frac{p}{p_{\max}}, 0, 1\right), \quad \hat{\dot{p}} = \text{clip}\left(\frac{\max(0, \dot{p})}{\dot{p}_{\max}}, 0, 1\right) \quad (1)$$

Time arguments (t) are omitted for brevity. Rubbing may use a friction proxy $\hat{f}$.

A.2 Rubbing Model

Parameters & Features. Rubbing presets $m_{\text{rub}} = \{R_{\text{mi}}, G_{\text{ma}}, H_{\text{ha}}, S_{\text{ss}}\}$ control roughness, granularity, resonance, and stick-slip. Features are $a_{\text{rub}} = \{v, p\}$.

Synthesis Structure. Rubbing combines a friction bed, granular events, and stick-slip, followed by resonance h_{rub} :

$$y_{\text{rub}} = g_{\text{rub}}(y_{\text{bed}} + y_{\text{grain}} + y_{\text{ss}}) * h_{\text{rub}}(m_{\text{rub}}) \quad (2)$$

Friction Bed. Colored noise $n(t)$ is modulated by amplitude A_{bed} and bandwidth:

$$y_{\text{bed}} = A_{\text{bed}}(1 + d_{\text{am}}u)\text{BPF}(n; f_c, Q) \quad (3)$$

The friction-power proxy e and amplitude A_{bed} (with compression) are:

$$e = \hat{v}\hat{p}, \quad A_{\text{bed}} = \alpha_0 + \alpha_1 \log(\epsilon + e) \quad (4)$$

Brightness f_c and modulation depth d_{am} are controlled by:

$$f_c = f_{\min} \exp\left(\hat{v} \ln \frac{f_{\max}}{f_{\min}}\right), \quad d_{\text{am}} = \beta_0 + \beta_v \hat{v} + \beta_R R_{\text{mi}} \quad (5)$$

Granular Events. Discrete impacts follow a Poisson process with rate λ_{gr} and burst $b(t)$:

$$y_{\text{gr}} = \sum_i A_i b(t - t_i), \quad A_i \propto \hat{p}^\gamma \text{imp}(G_{\text{ma}}), \quad \lambda_{\text{gr}} = \lambda_0 + \lambda_1 \hat{v}^k \text{den}(G_{\text{ma}}) \quad (6)$$

Stick-slip. When enabled ($S_{\text{ss}} > 0$), a speed-dependent pulse train is added:

$$y_{\text{ss}} = S_{\text{ss}} A_{\text{ss}} \text{Res}\left(\sum_j \delta(t - t_j); f_{\text{ss}}, Q_{\text{ss}}\right) \quad (7)$$

A.3 Pressing Model

Parameters & Gating. $m_{\text{pr}} = \{G_{\text{si}}, K_{\text{co}}, H_{\text{gr}}\}$ control grain size, compaction, and hardness. Sounds are suppressed during tangential motion.

Compaction State. To model material “packing,” state $c \in [0, 1]$ tracks press and release:

$$c(t + \Delta t) = \text{clip}(c + \kappa_{\uparrow} \hat{p} \Delta t - \kappa_{\downarrow} \hat{p}_{\text{rel}} \Delta t, 0, 1) \quad (8)$$

Micro-crunch events. Event rate λ_{pr} decreases as compaction c saturates:

$$\lambda_{\text{pr}} = \Lambda \hat{p}^k (1 - c)^q (0.25 + 0.75 K_{\text{co}})(0.15 + 0.85(1 - G_{\text{si}})) \quad (9)$$

The resulting sound y_{pr} uses bursts b_{pr} with amplitude A_i scaled by parameters:

$$y_{\text{pr}} = \sum_i A_i b_{\text{pr}}, \quad A_i \propto \hat{p}^\eta \prod \text{params}(G_{\text{si}}, H_{\text{gr}}, 1 - c) \quad (10)$$

A.4 Tapping Model

Parameters. $m_{\text{tap}} = \{H_{\text{ta}}, D_{\text{da}}, T_{\text{to}}, C_{\text{ca}}\}$ control hardness, decay, tonality, and hollowness.

Synthesis Structure. Tapping uses an attack transient, modal bank, and tail:

$$y_{\text{tap}} = g_{\text{tap}}(y_{\text{atk}} + y_{\text{modes}} + y_{\text{tail}}) \quad (11)$$

Overall gain g_{tap} is mapped from normalized peak pressure $\hat{p}_{\text{pk}}$:

$$\hat{p}_{\text{pk}} = \text{clip}(p_{\text{pk}}/p_{\max}, 0, 1), \quad g_{\text{tap}} = (\epsilon + \hat{p}_{\text{pk}})^\alpha \quad (12)$$

Attack Transient. A short noise burst $w(t)$ biased by hardness and strike rate $\hat{p}_{\text{pk}}$:

$$y_{\text{atk}} = A_{\text{atk}} w(t) e^{-t/\tau_{\text{atk}}}, \quad A_{\text{atk}} \propto 1 + \rho_H H_{\text{ta}} + \rho_r \hat{p}_{\text{pk}} \quad (13)$$

Modal Bank. A sum of N decaying sinusoids:

$$y_{\text{modes}} = \sum_{i=1}^N A_i \sin(2\pi f_i t + \phi_i) e^{-t/\tau_i} \quad (14)$$

Decay times τ_i are shortened by material damping and contact duration $contact_s$:

$$\tau_i = \frac{\tau_{i,0}(D_{\text{da}})}{1 + \rho_c contact_s} \quad (15)$$